

Fractions, Decimals and Percentages — Paper C

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Anything you cannot do in your head, do on paper.
- Give fractions in their simplest form unless the question says otherwise.
- The mark for each question is shown in square brackets on the right.
- Two questions on this paper have no obvious first step. If one stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – DECIMALS AND REVERSE PERCENTAGES

- 1.** For each fraction, state whether its decimal **terminates** or **recurs**, and write the decimal.

(a) $3/16$ [1]

(b) $5/12$ [1]

(c) $7/25$ [1]

You can tell before dividing. Look at the prime factors of the denominator.

- 2.** After a 20% increase, the price of a book is 96 rupees. Find the price before the increase. [3]

3. A price is reduced by 25% to 180 rupees. Find the original price. [3]

SECTION 2 – SUCCESSIVE CHANGE

4. A sum of money is increased by 10%, and the result is then decreased by 10%.
(a) Show, using a starting value of your choice, that the final amount is not the same as the original. [2]

- (b) State the overall percentage change. [1]

5. Work out $\frac{2}{3}$ of $\frac{3}{5}$ of 450. [3]

6. A jacket has a marked price of 2400 rupees. Two shops sell it.
Shop A: a straight 30% off.
Shop B: $\frac{1}{4}$ off the marked price, then a further 10% off that reduced price.

- (a) Find the price at each shop. [2]

- (b) State which shop is cheaper, and by how much. [2]

7. A number is $\frac{5}{8}$ of 640. Find 15% of that number.

[3]

8. A bag of rice loses 12% of its mass. It now has a mass of 4.4 kg. Find the original mass of the bag.

[3]

9. Write the recurring decimal $0.4444\dots$ as a fraction in its simplest form.

[2]

The recurring block is one digit long. Nines are worth knowing by heart.

10. In a sale a coat is reduced by 35%, and then a further 10% is taken off at the till. A sign says “**45% off everything**”. Explain why the sign is wrong, and give the correct single percentage discount.

[3]

End of paper. Check every answer has a unit or a form the question actually asked for.